Python Set Data Type – Complete Notes

Definition: A set is a built-in Python data type used to store multiple unique values. A set is an unordered collection of unique elements.

1. Creating a Set

```
s = {10, 20, 30, 40}
print(s)
```

2. Empty Set

```
s = set()
print(s)

# Note: {} creates an empty dictionary, not an empty set.
```

3. Important Properties of Set

• Unordered

• Duplicate values are not allowed

• Mutable

• No indexing

• No slicing

• Can contain different immutable data types

• Mutable elements such as lists cannot be stored

• Empty set is created using set()

4. Duplicate Values

```
s = {10, 20, 10, 30, 20}
print(s)

# Output: {10, 20, 30}
```

5. Indexing is Not Supported

```
s = {10, 20, 30}

# s[0] # Error

for x in s:
print(x)
```

6. Adding One Element – add()

```
s = {10, 20, 30}
s.add(40)
print(s)
```

7. Adding Multiple Elements – update()

```
s = {10, 20}
s.update([30, 40, 50])
print(s)
```

8. Removing – remove()

```
s = {10, 20, 30}
s.remove(20)
print(s)

# Error occurs if 20 is not present.
```

9. Removing – discard()

```
s = {10, 20, 30}
s.discard(20)
print(s)

# No error if the element is absent.
```

10. pop()

```
s = {10, 20, 30}
x = s.pop()
print(x)
print(s)

# pop() removes an arbitrary element.
```

11. clear()

```
s = {10, 20, 30}
s.clear()
print(s)

# Output: set()
```

12. del

```
s = {10, 20, 30}
del s

# The complete set variable is deleted.
```

13. Union

```
a = {1, 2, 3, 4}
b = {3, 4, 5, 6}

print(a | b)
print(a.union(b))

# Output: {1, 2, 3, 4, 5, 6}
```

14. Intersection

```
a = {1, 2, 3, 4}
b = {3, 4, 5, 6}
```

```
print(a & b)
print(a.intersection(b))

# Output: {3, 4}
```

15. Difference

```
a = {1, 2, 3, 4}
b = {3, 4, 5, 6}

print(a - b)
print(a.difference(b))

# Output: {1, 2}
```

16. Symmetric Difference

```
a = {1, 2, 3, 4}
b = {3, 4, 5, 6}

print(a ^ b)
print(a.symmetric_difference(b))

# Output: {1, 2, 5, 6}
```

17. Membership Operators

```
s = {10, 20, 30}

print(20 in s)
print(50 in s)
print(50 not in s)
```

18. Set Comparison

```
a = {1, 2}
b = {1, 2, 3, 4}

print(a.issubset(b))
print(a <= b)

print(b.issuperset(a))
print(b >= a)
```

19. Disjoint Set

```
a = {1, 2, 3}
b = {4, 5, 6}

print(a.isdisjoint(b))

# Output: True
```

20. Length of Set

```
s = {10, 20, 30, 40}
print(len(s))
```

```
# Duplicate values are counted only once.
```

21. Convert List to Set

```
numbers = [10, 20, 10, 30, 20, 40]
s = set(numbers)
print(s)

# Useful for removing duplicates.
```

22. Convert Set to List

```
s = {10, 20, 30}
numbers = list(s)
print(numbers)
```

Set Methods – Quick Reference

Method	Use
add()	Add one element
update()	Add multiple elements
remove()	Remove element; error if absent
discard()	Remove element; no error if absent
pop()	Remove an arbitrary element
clear()	Remove all elements
union()	Combine sets
intersection()	Find common elements
difference()	Find elements only in first set
symmetric_difference()	Find non-common elements
issubset()	Check subset
issuperset()	Check superset
isdisjoint()	Check whether sets have no common elements
copy()	Create a copy

Set Operators

Operator	Meaning
|	Union
&	Intersection
-	Difference
^	Symmetric Difference
<=	Subset
>=	Superset
in	Membership
not in	Not membership

7 Simple Set Programs

Example 1 – Create and Print

```
numbers = {10, 20, 30, 40, 50}
print(numbers)
```

Example 2 – Remove Duplicates

```
numbers = {10, 20, 10, 30, 20, 40}
print(numbers)
```

Example 3 – Add Element

```
numbers = {10, 20, 30}
numbers.add(40)
print(numbers)
```

Example 4 – Remove Element

```
numbers = {10, 20, 30, 40}
numbers.remove(30)
print(numbers)
```

Example 5 – Common Elements

```
a = {10, 20, 30}
b = {20, 30, 40}
print(a & b)
```

Example 6 – Union

```
a = {10, 20, 30}
b = {30, 40, 50}
print(a | b)
```

Example 7 – Even Numbers

```
numbers = {10, 15, 20, 25, 30, 35}

for x in numbers:
if x % 2 == 0:
print(x)
```

Quick Revision

Set = Unique + Unordered + Mutable + No Indexing + No Slicing.

Created with: {1, 2, 3}
Empty set: set()
Duplicates are automatically removed.
Most important methods: add(), update(), remove(), discard(), pop(), clear(), union(), intersection(), difference(), symmetric_difference(), issubset(), issuperset(), isdisjoint().

Remember: remove() gives an error when the element is absent, while discard() does not.